Data Science Job-Ready Sprint — 20 hrs/week

Long arc: Full roadmap coverage in ~~6–8 months at 20 hrs/week (~~500–650 hrs total). **Sprint (this doc):** Weeks 1–13 → *application-ready*. Then months 4–7 = mastery tail.

Target for Month 3: Fluent on fundamentals, 2 portfolio projects live on GitHub, resume + LinkedIn ready, and interview-capable on SQL, stats, and core ML. **You start applying at Week 12.**

Weekly rhythm (how to spend the 20 hrs)

- **~12 hrs** — new material: read/watch, then immediately code it yourself.
- **~5 hrs** — hands-on project work and exercises.
- **~3 hrs** — review, quizzes, spaced repetition, daily SQL drills.

Gating rule: At the end of each stage I quiz you. You don't advance until you demonstrate understanding — honestly assessed, gaps named.

PHASE 1 — The 13-Week Sprint (Application-Ready)

Weeks 1–2 · Programming Foundations + Data Wrangling (~40 hrs)

Roadmap: 01 Programming Foundations, 03 Data Manipulation

- Python refresh (fast — you're rusty, not blank): data types, control flow, functions, comprehensions, modules, virtual environments, Git basics.
- NumPy + pandas: Series, DataFrames, indexing, filtering, groupby.
- Data cleaning: missing values, duplicates, outliers, type conversion, categorical encoding, feature scaling.
- **Deliverable:** a clean, well-commented wrangling notebook on a real messy dataset (UCI or Kaggle).

Weeks 3–4 · SQL + Databases (~40 hrs)

Roadmap: 05 Database Management

- Relational: schema design, SELECT, WHERE, JOINs (all types), aggregation, GROUP BY, subqueries, window functions, indexing.
- NoSQL: awareness only (document vs key-value vs graph, when each is used).
- **Daily 20-min SQL drill** — this is the most common first interview screen, so we drill it to reflex.
- **Deliverable:** a solved SQL problem set + one real analytical query project.

Weeks 5–6 · Statistics + EDA + Visualization (~40 hrs)

Roadmap: 02 Calculus & Statistics, 04 Data Visualization

- Probability distributions, hypothesis testing, p-values, confidence intervals, correlation vs causation, regression (theory).
- EDA workflow; matplotlib + seaborn; storytelling, annotation, audience targeting.
- **Deliverable → Portfolio Project 1:** an EDA + statistical analysis with a clear written narrative and polished visuals. *This is a "communicate insight" project.*

Weeks 7–9 · Core Machine Learning + Evaluation (~60 hrs)

Roadmap: 06 Machine Learning Basics, 07 Model Evaluation

- Supervised: linear & logistic regression → decision trees → random forests → gradient boosting; KNN & SVM (lighter).
- Unsupervised (lighter): k-means, PCA.
- Feature engineering woven throughout.
- Evaluation rigor: train/test split, cross-validation, precision/recall/F1/AUC, MSE/MAE, bias–variance, overfitting, regularization, hyperparameter tuning.
- **Deliverable → Portfolio Project 2:** a full end-to-end supervised ML project — problem framing → sourcing → cleaning → modeling → evaluation → results communication.

Weeks 10–11 · Specialization + DL concepts + Deployment (~40 hrs)

Roadmap: 08 NLP, 09 Deep Learning (concepts), 10 Deployment

- One specialization — **NLP** (tokenization, embeddings, sentiment) given rising market demand; swap for time series if your target roles skew that way.
- Deep learning: *concepts only* — neural nets, activations, forward/backprop, gradient descent, loss functions. Small demo, not mastery.
- Deploy one model (Streamlit or FastAPI) so a project is clickable end-to-end.

Weeks 12–13 · Interview Prep + Polish + Apply (~40 hrs)

Roadmap: 12 Professional Development

- SQL drills, stats/ML Q&A, coding practice, take-home simulation, behavioral/case questions, **mock interviews** on request.
- GitHub README polish, resume, LinkedIn, and your career-switch narrative.
- **Start applying (Week 12).** Applications run in parallel with continued learning.

PHASE 2 — Mastery Tail (Months 4–7, while applying/employed)

Depth on the heavy stages no junior is expected to have mastered up front:

- **Deep Learning:** CNNs, RNNs/LSTM/GRU, Transformers & attention, GANs.
- **Advanced Analytics:** full time series (stationarity, ARIMA, seasonality, forecasting), deeper NLP, topic modeling.
- **Big Data Engineering:** distributed computing, MapReduce, Spark, stream vs batch, data lakes, query optimization.
- **MLOps/Deployment depth:** containerization, orchestration, IaC, monitoring, deployment pipelines.
- **Professional/Ethics:** bias detection, data privacy, fairness, reproducibility.

Portfolio target by Week 13

1. **EDA + statistical storytelling** project (Wk 5–6)
2. **End-to-end supervised ML** project, deployed (Wk 7–11)
3. *(Optional stretch)* SQL analysis project (Wk 3–4)

Two strong, well-documented projects beat five half-finished ones. Quality and a clear README win interviews.